



**DUBLIN CITY UNIVERSITY**

## **SEMESTER 1 EXAMINATIONS 2016/2017**

**MODULE:** CA208 - Logic

**PROGRAMME(S):**

CASE - BSc in Computer Applications (Sft.Eng.)  
CPSSD - BSc in Computational Problem Solving and SW Dev.

**YEAR OF STUDY:** 2

**EXAMINERS:** David Sinclair (Ph:5510)

**TIME ALLOWED:** 2 hours

**INSTRUCTIONS:** Answer ALL 6 questions from Section A. Answer 2 out of 3 questions from Section B. Question 1 carries 30 marks. Questions 2 to 9 carry 10 marks.

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**PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO**

The use of programmable or text storing calculators is expressly forbidden.  
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

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*Requirements for this paper (Please mark (X) as appropriate)*

|                          |                           |
|--------------------------|---------------------------|
| <input type="checkbox"/> | <i>Log Tables</i>         |
| <input type="checkbox"/> | <i>Graph Paper</i>        |
| <input type="checkbox"/> | <i>Dictionaries</i>       |
| <input type="checkbox"/> | <i>Statistical Tables</i> |

|                          |                                  |
|--------------------------|----------------------------------|
| <input type="checkbox"/> | <i>Thermodynamic Tables</i>      |
| <input type="checkbox"/> | <i>Actuarial Tables</i>          |
| <input type="checkbox"/> | <i>MCQ Only - Do not publish</i> |
| <input type="checkbox"/> | <i>Attached Answer Sheet</i>     |

## Section A

### QUESTION 1

[Total marks: 30]

Every lion chases some gazelle. Every gazelle who is old is smart. No lion catches any smart gazelle. Any lion who chases some gazelle but does not catch it is hungry.

1(a) [10 Marks]

Convert the statements above into a set of predicate formulae.

1(b) [10 Marks]

Convert the predicate formulae from Question 1(a) into *Prenex Conjunctive Normal Form (PCNF)*.

1(c) [10 Marks]

Using General Resolution, prove that if all gazelles are old then all lions are hungry.

[End Question 1]

### QUESTION 2

[Total marks: 10]

2(a) [10 Marks]

Show that

$$(A \vee B) \rightarrow C \equiv (A \rightarrow C) \wedge (B \rightarrow C)$$

[End Question 2]

### QUESTION 3

[Total marks: 10]

[10 Marks]

Using *semantic tableaux* prove the following proposition is valid.

$$\neg(p \wedge q) \leftrightarrow \neg p \vee \neg q$$

[End Question 3]

**QUESTION 4****[Total marks: 10]**

[10 Marks]

Using Gentzen system  $\mathcal{G}$  prove

$$(p \rightarrow q) \rightarrow ((p \rightarrow \neg q) \rightarrow \neg p)$$

**[End Question 4]****QUESTION 5****[Total marks: 10]**

[10 Marks]

Using *semantic tableaux* prove the following first-order statement is valid.

$$\vdash (\forall x.p(x) \vee \forall x.q(x)) \rightarrow \forall x.(p(x) \vee q(x))$$

**[End Question 5]****QUESTION 6****[Total marks: 10]**

[10 Marks]

Using *Skolemisation*, convert the following predicate expression into a set of clauses.

$$\exists y.(\exists x.Q(x, y) \vee \exists x.P(x)) \rightarrow \neg(\neg\exists x.P(x) \wedge \forall x.\exists y.Q(y, x))$$

**[End Question 6]**

## Section B

### QUESTION 7

**[Total marks: 10]**

[10 Marks]

Given the following Prolog relations:

- `parent(X,Y)` is true if X is the parent of Y,
- `male(X)` is true if X is male, and
- `female(X)` is true if X is female,

write the following Prolog relations:

- `grandmother(X,Y)` is true if X is the grandmother of Y,
- `aunt(X,Y)` is true if X is an aunt of Y,
- `ancestor(X,Y)` is true if X is a direct ancestor of Y.

**[End Question 7]**

### QUESTION 8

**[Total marks: 10]**

[10 Marks]

Given the relation `road(A,B,N)` is true if a one-way road of length N km exists from A to B and assuming that the set of `road` relations does not contain any loops where it is possible to travel from any town back to the same town, write the following Prolog relations:

- `routen(X,Y,N)` is true if there exists a route of length N, with zero or more intermediate towns, between X and Y.
- `toofar(X,Y)` is true if there is a route between X and Y and the total distance is more than 6 km.

**[End Question 8]**

### QUESTION 9

**[Total marks: 10]**

[10 Marks]

Describe what *negation as failure* means in Prolog. Why is negation problematical in Prolog? Illustrate your answer with examples.

**[End Question 9]**

## [APPENDICES]

### Semantic Tableaux

| $\alpha$                    | $\alpha_1$            | $\alpha_2$            | $\beta$                         | $\beta_1$                   | $\beta_2$                   |
|-----------------------------|-----------------------|-----------------------|---------------------------------|-----------------------------|-----------------------------|
| $\neg\neg A_1$              | $A_1$                 |                       | $\neg(B_1 \wedge B_2)$          | $\neg B_1$                  | $\neg B_2$                  |
| $A_1 \wedge A_2$            | $A_1$                 | $A_2$                 | $B_1 \vee B_2$                  | $B_1$                       | $B_2$                       |
| $\neg(A_1 \vee A_2)$        | $\neg A_1$            | $\neg A_2$            | $B_1 \rightarrow B_2$           | $\neg B_1$                  | $B_2$                       |
| $\neg(A_1 \rightarrow A_2)$ | $A_1$                 | $\neg A_2$            | $B_1 \uparrow B_2$              | $\neg B_1$                  | $\neg B_2$                  |
| $\neg(A_1 \uparrow A_2)$    | $A_1$                 | $A_2$                 | $\neg(B_1 \downarrow B_2)$      | $B_1$                       | $B_2$                       |
| $A_1 \downarrow A_2$        | $\neg A_1$            | $\neg A_2$            | $\neg(B_1 \leftrightarrow B_2)$ | $\neg(B_1 \rightarrow B_2)$ | $\neg(B_2 \rightarrow B_1)$ |
| $A_1 \leftrightarrow A_2$   | $A_1 \rightarrow A_2$ | $A_2 \rightarrow A_1$ | $B_1 \oplus B_2$                | $\neg(B_1 \rightarrow B_2)$ | $\neg(B_2 \rightarrow B_1)$ |
| $\neg(A_1 \oplus A_2)$      | $A_1 \rightarrow A_2$ | $A_2 \rightarrow A_1$ |                                 |                             |                             |

  

| $\gamma$             | $\gamma(a)$ | $\delta$             | $\delta(a)$ |
|----------------------|-------------|----------------------|-------------|
| $\forall x.A(x)$     | $A(a)$      | $\exists x.A(x)$     | $A(a)$      |
| $\neg\exists x.A(x)$ | $\neg A(a)$ | $\neg\forall x.A(x)$ | $\neg A(a)$ |

### Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$  is a *sequent* and means that if  $A$  is true, then  $B$  can be inferred.

| $\alpha$                        | $\alpha_1$                  | $\alpha_2$                  | $\beta$                     | $\beta_1$             | $\beta_2$             |
|---------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------|-----------------------|
| $\neg\neg A$                    | $A$                         |                             | $B_1 \wedge B_2$            | $B_1$                 | $B_2$                 |
| $\neg(A_1 \wedge A_2)$          | $\neg A_1$                  | $\neg A_2$                  | $\neg(B_1 \vee B_2)$        | $\neg B_1$            | $\neg B_2$            |
| $A_1 \vee A_2$                  | $A_1$                       | $A_2$                       | $\neg(B_1 \rightarrow B_2)$ | $B_1$                 | $\neg B_2$            |
| $A_1 \rightarrow A_2$           | $\neg A_1$                  | $A_2$                       | $\neg(B_1 \uparrow B_2)$    | $B_1$                 | $B_2$                 |
| $A_1 \uparrow A_2$              | $\neg A_1$                  | $\neg A_2$                  | $B_1 \downarrow B_2$        | $\neg B_1$            | $\neg B_2$            |
| $\neg(A_1 \downarrow A_2)$      | $A_1$                       | $A_2$                       | $B_1 \leftrightarrow B_2$   | $B_1 \rightarrow B_2$ | $B_2 \rightarrow B_1$ |
| $\neg(A_1 \leftrightarrow A_2)$ | $\neg(A_1 \rightarrow A_2)$ | $\neg(A_2 \rightarrow A_1)$ | $\neg(B_1 \oplus B_2)$      | $B_1 \rightarrow B_2$ | $B_2 \rightarrow B_1$ |
| $A_1 \oplus A_2$                | $\neg(A_1 \rightarrow A_2)$ | $\neg(A_2 \rightarrow A_1)$ |                             |                       |                       |

  

| $\gamma$             | $\gamma(a)$ | $\delta$             | $\delta(a)$ |
|----------------------|-------------|----------------------|-------------|
| $\exists x.A(x)$     | $A(a)$      | $\forall x.A(x)$     | $A(a)$      |
| $\neg\forall x.A(x)$ | $\neg A(a)$ | $\neg\exists x.A(x)$ | $\neg A(a)$ |

[END OF APPENDICES]

[END OF EXAM]